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Paper Code : PCC- CS401/PCC-CS401/PCCCS 401/PCCCS401 Discrete Mathematics

UPID : 004407

Time Allotted : 3 Hours

Full Marks :70

*The Figures in the margin indicate full marks.**Candidate are required to give their answers in their own words as far as practicable***Group-A (Very Short Answer Type Question)**

1. Answer any ten of the following :

[1 x 10 = 10]

- (i) Find the domain of $f(x)$ where $f(x) = \sqrt{\log \frac{4x-x^2}{3}}$, $x, f(x)$ are real.
- (ii) What is the rank of the word "SACHIN" in dictionary wise?
- (iii) Write the CNF form of $(p \rightarrow (q \wedge \neg r))$.
- (iv) State Lagrange's theorem on group.
- (v) Find the minimum and maximum number of edges of a simple graph with 10 vertices and 3 components.
- (vi) If $aN = \{ax : x \in N\}$ then $7N \cap 3N =$
- (vii) How many ways are there to travel in x-y-z space from the origin (0,0,0) to the point (4,3,5) by taking unit steps in positive x,y,z directions only?
- (viii) Express the proposition $p \leftrightarrow q$ in the minimal complete set of connectives $\{\neg, \vee\}$.
- (ix) Does the ring Z_6 of integers modulo 6 is integral domain? Justify.
- (x) Define separable graph.
- (xi) Define enumerable set and give an example.
- (xii) How many square are there in chess board?

Group-B (Short Answer Type Question)

Answer any three of the following :

[5 x 3 = 15]

2. Show that a cycle with n vertices is a perfect graph if n is even. [5]
3. Let D_{36} be the set of all positive divisors of 36. Show that D_{36} is a POSET w.r.t. the relation 'divisor'. Draw the Hasse diagram of this POSET. Find the Supremum and Infimum of the set $\{4, 9\}$. [5]
4. How many eight bit binary strings are possible that never begin either with 011 or with 101? [5]
5. Check the validity of the following argument : [5]
"If my program runs successfully, then I will submit my project. I can appear in the examination only if I submit my project. Either my program runs successfully or the computer crashes. Therefore, if the computer does not crash then I can appear at the examination".
6. Find an equivalent expression for $p \wedge (q \leftrightarrow r) \vee (r \leftrightarrow p)$ that contains minimal complete set of connectives. [5]

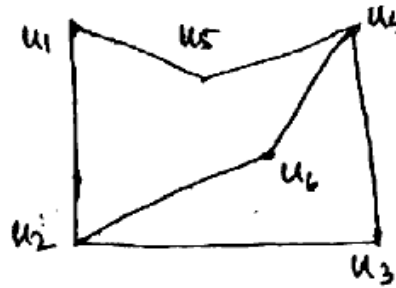
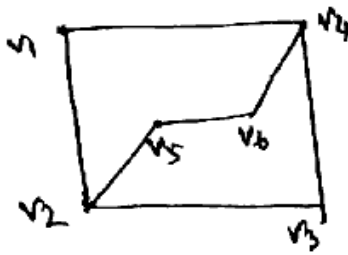
Group-C (Long Answer Type Question)

Answer any three of the following :

[15 x 3 = 45]

7. (a) Show that a simple graph with n number of vertices and k number of components can have at most $(n-k)(n-k+1)/2$ edges. [5]
- (b) Show the following: [5]

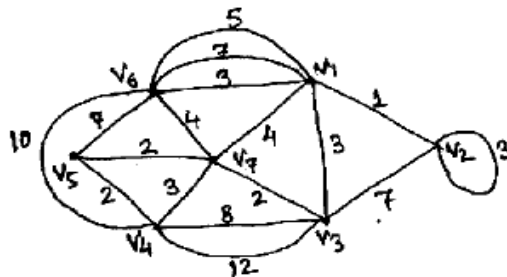
Show that the following two graphs are isomorphic:-



(c) Solve the problem:

[5]

By Dijkstra's Algorithm find the shortest path and shortest distance from vertex v_2 to vertex v_5 in the following graph.



8. (a) Find the total number of integers between 1 and 1000 which are neither perfect square nor perfect cubes? [5]
- (b) Find the number of ways a company can assign 7 projects to 4 people so that each person gets at least one project. [5]
- (c) Consider 6 people. Any two of them are either friends or strangers. Find the number of selected persons where the persons are either mutual friends or mutual strangers. Explain the method. [5]
9. (a) Find all solutions in positive integers of $5x + 3y = 52$. [5]
- (b) If $\gcd(a, b) = 1$, prove that $\gcd(a + b, a^2 - ab + b^2) = 1$ or 3. [5]
- (c) Show that the product of any r consecutive natural numbers is divisible by $r!$ [5]
10. (a) In an College offering a course in which every student will have to take English and one of the following groups (Botany, Zoology, Microbiology), (Botany, Chemistry, Zoology), (Physics, Zoology, Botany), (Maths, Physics, Chemistry), (Maths, Physics, Statistics), (Maths, Statistics, Computer), (Computer, Chemistry, Microbiology). Find the period allocation in a day such that students can attend their classes without any hindrance. [5]
- (b) Show that s is a valid conclusion from the premises: $p \rightarrow \neg q, q \vee r, \neg s \rightarrow p, \neg r$ [5]
- (c) Write the converse, inverse and contrapositive of $(p \wedge q) \rightarrow \neg q$. [5]
- Also write the contrapositive statement of the statement: "If a triangle is right angled then it is not equilateral"
11. (a) [5]

Let $GL(2, \mathbb{R})$ be the multiplicative group of all real non singular matrices of order 2. Show that the set

$$H = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : ad - bc = 1 \right\} \text{ is a subgroup of } GL(2, \mathbb{R}).$$

- (b) Show that the set G of all ordered pairs (a, b) with $a \neq 0$, of real numbers a, b forms a group with operation $\circ$ defined by [5]
 $(a, b) \circ (c, d) = (ac, bc + d)$
- (c) Let G be an abelian group. Prove that the subset $S = \{p \in G : p = p^{-1}\}$ forms a subgroup of G . [5]

*** END OF PAPER ***